

Integrative Learning Strategies for Manufacturing Time Estimation Across Diverse Product Families

Le Vi Nhan Tam Tran

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² Supervisor(s) : Abdourahim Sylla, Bruno Agard

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Abstract : *Accurate estimation of manufacturing time is crucial in Engineer-To-Order (ETO) industries in order to optimize workflows, reduce costs and improve overall efficiency. This study presents an integrative machine learning framework that integrates supervised, unsupervised and memory-based learning techniques to enhance the accuracy of estimation. This approach contributes to two key areas : (i) identifying critical features for time estimation and automating product family classification using Rule-Based Learning (RBL) and (ii) improving case retrieval in Case-Based Reasoning (CBR) by leveraging a KD-Tree structure for more efficient searches. Notably, the proposed method effectively handles highly variable, multi-family products within a single, scalable framework. The model achieves a 10% reduction in Mean Absolute Error (MAE) and a 60% decrease in computational cost, significantly accelerating retrieval speed. Validated on real-world manufacturing data from a French metallurgy company, the approach demonstrates superior accuracy and efficiency, reinforcing its suitability for complex Engineering-To-Orders environments.*

Keywords : *Engineer-To-Order (ETO), Manufacturing Time, Product family, Case-Based Reasoning (CBR), Rule-Based Learning (RBL), Machine Learning (ML).*

1 Introduction

Engineer-to-Order (ETO) manufacturing is distinguished by its high degree of product customization, where each item is uniquely crafted to meet a client's specific technical and functional requirements (Sylla et al). While this tailored production model offers flexibility and competitive value, it also introduces substantial challenges—particularly when attempting to accurately estimate manufacturing time (Chehade et al). Accurate manufacturing time estimation is critical in industrial settings, as it directly influences production planning, resource allocation, cost control, and the ability to meet customer deadlines, ultimately contributing to more efficient workflows and improved operational performance. However, the diversity in product specifications, from geometrical features to material compositions and process constraints, creates a constantly shifting estimation landscape. This variability demands estimation methods that are not only accurate but also adaptable and responsive to real-time complexity.

A core difficulty arises from the inherent variability across product families within ETO environments. Each family is characterized by unique combinations of features and manufacturing rules, yet many predictive models in the literature tend to treat all products homogeneously (Perez et al; Li and Lee). This oversimplification limits their ability to capture the subtle but critical distinctions between product groups, thereby reducing both prediction accuracy and model generalizability. Compounding this issue, the inability to properly distinguish product families exacerbates challenges in retrieving relevant historical data for prediction tasks.

Machine learning has become a widely adopted approach for manufacturing time estimation due to its ability to capture complex, non-linear relationships from large datasets. However, these models often require large volumes of high-quality, labeled data and can struggle to adapt to the inherent variability and heterogeneity of ETO environments. In parallel, the efficiency and scalability of estimation methods based on Case-Based Reasoning (CBR) are significantly challenged by the growing volume of historical cases, particularly in high-dimensional feature spaces. Naive retrieval strategies, such as exhaustive similarity searches, become computationally intensive and increasingly ineffective as dataset size and feature complexity increase (Aamodt et al; Kim and Han). Without intelligent mechanisms to narrow the search space—such as family-specific clustering or spatial indexing—retrieving the most relevant examples becomes slow and prone to errors, compromising prediction accuracy (Mansoul and Atmani; Müller and Bergmann). Consequently, scalable and precise retrieval techniques, such as KD-Trees or hierarchical clustering, are essential to maintain both computational efficiency and prediction quality in complex manufacturing environments (Zhang et al).

Crucially, the robustness of manufacturing time estimation hinges on the quality of input features, as real-world manufacturing datasets are often marred by missing entries, inconsistent scales, and qualitative descriptors that diminish modeling value if not properly addressed (Omuya et al). Effective feature engineering—encompassing feature scaling, categorical encoding, dimensionality reduction, and similarity-based weighting—is a critical yet often underexplored step in enhancing predictive performance (He et al). By systematically transforming raw inputs into structured, informative representations, these steps strengthen the model’s resilience to noise and variability—critical for achieving accurate and generalizable manufacturing time estimations in complex ETO.

Addressing these interconnected challenges requires an integrative learning framework—one that strategically combines targeted product family classification, efficient case retrieval and similarity-based reasoning. By integrating multiple AI techniques into a cohesive estimation chain, we can build solutions that are not only accurate, but also scalable and adaptive to the complexity inherent in Engineer-to-Order (ETO) manufacturing environments. In this work, we propose a new integrative machine learning framework designed to provide a comprehensive solution for manufacturing time estimation in ETO settings.

The proposed framework is organized into three sequential phases to enhance manufacturing time estimation.

- Phase 1 : Product Family Classification begins with a new product characterized by a set of features such as geometry, material type, and production parameters. These features are standardized and then processed using either a Rule-Based Learning (RBL) or an Artificial Neural Network (ANN) model to determine which product family it belongs to. This classification ensures that the product is analyzed within the right context before moving to case retrieval.

- Phase 2 : Similar Case Retrieval via KD-Tree by using the classified product family and its key features, a KD-Tree structure is applied to efficiently retrieve the most relevant historical manufacturing cases - Relevant Case Set (RS). This helps narrow the search space and identify only the most relevant examples for estimation..
- Phase 3 : Manufacturing Time Estimation via CBR starting from the Relevant Case Set (RS) retrieved in Phase 2, CBR is applied to estimate the manufacturing time. The new product instance is compared to each case in the RS to compute similarity scores. These scores are then used to rank the top similar cases and select the Most Similar Cases (MSC). A weighted aggregation is performed on the manufacturing times of these MSCs, where each case contributes proportionally to its similarity score, resulting in the final estimated time for the new product.

The primary contribution of this study lies in the development of an integrated machine learning (ML) framework for manufacturing time estimation (MTE) tailored to diverse product families in Engineer-to-Order (ETO) environments. By combining product family classification—via Neural Networks or Rule-Based Learning—with a KD-Tree-enhanced Case-Based Reasoning (CBR) mechanism, the proposed approach effectively adapts to the structural variability and complexity inherent in real-world manufacturing data. On the practical side, this framework is validated through its application to a real industrial case from the French metallurgy sector. This industrial validation demonstrates not only the feasibility but also the efficiency and accuracy of the method, highlighting its potential as a scalable and data-driven solution for improving estimation performance in complex manufacturing settings.

This article begins in Section 1 with an overview of the scientific and industrial context, identifying the key challenges in manufacturing time estimation (MTE) within Engineer-to-Order (ETO) environments and presenting a general methodological approach to address these gaps. Section 2 with a review of related work, exploring how artificial intelligence techniques—ranging from machine learning to memory-based methods—have been applied to manufacturing time estimation and product family classification. We also examine clustering strategies for improving retrieval in CBR systems and conclude by identifying key research gaps, as formally outlined in Section 3. Section 4 introduces our proposed integrative framework, detailing the data preparation pipeline, the integration of rule-based and machine learning classifiers and the incorporation of KD-Tree structures to optimize the CBR process. Section 5 shifts from theory to practice, showcasing experimental results drawn from an industrial case study and evaluating the performance of our approach across multiple product families, with detailed discussion provided in Section 6. The article concludes in Section 7 by reflecting on the broader implications of our findings and outlining opportunities for future research and deployment.

2 State of the Art

This section reviews key Artificial Intelligence (AI) techniques applied to manufacturing time estimation and product family classification, with a focus on highly customized production settings. Section 2.1 covers machine learning, case-based reasoning (CBR), and hybrid models for time prediction. Section 2.2 discusses classification methods for grouping products into families, emphasizing neural networks and rule-based approaches. Section 2.3 explores clustering techniques used to enhance retrieval in Case-Based Reasoning.

2.1 Artificial Intelligence in Manufacturing Time Estimation

Machine Learning (ML) has emerged as a key approach for modeling complex, non-linear relationships in manufacturing data, enabling systems to learn from historical records and predict manufacturing times with improved accuracy. For instance, (Kumru and Kumru) demonstrated how Artificial Neural Networks (ANNs) outperformed traditional regression models in forecasting operation times in the metal industry. Similarly, (Liu et al) proposed a cluster-based evolving model for time estimation in semiconductor manufacturing, while (Hung) enhanced classification performance by incorporating rough set-based feature selection. These studies underline ML's ability to generalize across diverse datasets. More recent research has reinforced this potential. (Yamashiro & Nonaka) used LightGBM and Ridge regression models to estimate complex processing times in real-world factory data, achieving a 30% reduction in makespan when used for parallel machine scheduling—outperforming probabilistic approaches. Likewise, (Ma'ruf et al) showed that ANN models significantly outperformed CAM-based estimation tools in MTO environments by incorporating geometry and tolerance parameters often overlooked in standard software, achieving a 72% reduction in RMSE. These contributions highlight ML's adaptability and capacity to extract critical predictors from manufacturing data. Despite these successes, purely data-driven models often struggle with sparse or variable input dynamic dataset, a common feature in ETO production, where new product types and shifting specifications continuously reshape the data landscape.

In parallel, Case-Based Reasoning (CBR) offers an intuitive solution to manufacturing time prediction by drawing on past experiences to solve new problems. It is particularly suited to ETO settings where historical records of similar products can guide new estimations. For example, (Chang et al) used CBR in semiconductor manufacturing, utilizing historical flow times to improve timeline predictions. (Sylla et al) applied CBR in ETO systems to evaluate performance, confirming its relevance for custom orders. Knowledge-based extensions of CBR have also shown promise. For example, (Mourtzis et al) combined CBR with rule-based reasoning for estimating timelines of highly customized, engineer-to-order products. Additionally, (Dattani and Magaldi) demonstrated the value of CBR for decision-making in the aviation industry by integrating it with ML for time estimation. However, many existing CBR systems often lack retrieval and adaptation mechanisms, especially for real-world problems.

To overcome the limitations of standalone approaches, recent studies have focused on coupling ML and CBR. These combinations aim to harness the generalization power of ML and the contextual relevance of CBR. (Jung et al) integrated radial basis function networks with CBR to enhance manufacturing design and lead time estimation. More recently, (Chehade and Sylla) combined CatBoost and CBR in ETO environments, illustrating how ML can enhance adaptation phases in CBR.

Abdelwahed et al further demonstrated the integration of ML with KD-Tree structures to improve retrieval speed and scalability in high-dimensional datasets.

Machine Learning models and Case-Based Reasoning each bring valuable strengths to manufacturing time estimation. While these hybrid approaches mark significant progress, however, many existing solutions fall short when applied to the highly variable context of Engineer-to-Order (ETO) manufacturing. Specifically, they often lack adaptability to diverse product families, struggle with heterogeneous data types and fail to address the imbalance between abundant data in some product families and scarce data in others.

Moreover, few studies present end-to-end frameworks that integrate automated product classification, feature transformation and scalable retrieval mechanisms—elements that are essential for delivering accurate, efficient, and generalizable time estimation in real-world industrial settings.

Motivated by these limitations, our work proposes an integrative machine learning chain that integrates automated classification (via Rule-Based Learning), data processing and optimized case retrieval using KD-Tree-enhanced CBR. This combination is designed to improve both the accuracy and adaptability of manufacturing time estimation across multiple product families. A key component of this framework is the ability to automatically classify new products into the appropriate product families—an essential step that ensures contextual relevance for downstream similarity-based reasoning. The following section explores existing AI techniques used for PF classification.

2.2 Artificial Intelligence in Product Families Classification

Accurate classification ensures that manufacturing time estimation models operate within homogeneous subsets of data, thereby enhancing both predictive accuracy and interpretability. Given the centrality of this task, it is critical to review existing artificial intelligence (AI) approaches

(Hung) introduced a hybrid neural architecture integrating Rough Set Attribute Reduction (RSAR) and Backpropagation Neural Network (BPNN) for the classification of diverse Integrated Circuit packaging types. The use of RSAR reduced dimensionality and removed redundant features, significantly boosting classification accuracy and computational efficiency. The RSAR-BPNN framework achieved over 98% test accuracy and highlighted the model’s applicability in high-dimensional, multi product families industrial scenarios—especially where fine-grained distinctions between similar product types are needed. In a more recent context, (Shad et al) applied neural networks to tackle a classification task complicated by vast item catalogs and unstructured textual data. They proposed two distinct architectures : a FastText-based extreme classification model capable of scaling to millions of product categories using hierarchical softmax, and a Siamese LSTM-based similarity model that effectively captured semantic relationships between seller listings and catalog items. Similarly, (Roberson) explored automated product categorization in the context of the U.S. Economic Census. By leveraging textual descriptions and Universal Product Codes (UPCs) from businesses, the study applied natural language processing (NLP) and machine learning models—such as Logistic Regression (LR), Multinomial Naive Bayes (MNB), and Support Vector Machines (SVM)—to classify products under the North American Product Classification System (NAPCS). SVM models achieved over 95% classification accuracy, demonstrating the feasibility of using lightweight, interpretable ML techniques for large-scale product classification. This work emphasizes the practicality and scalability of ML-based product classification

Traditional approaches to product family identification have long relied on rule-based learning to establish structured relationships between customer requirements and design parameters. These methods often depend heavily on expert-crafted configuration rules, which can be take time and may lack scalability. For example, (Zhu and Jiang) introduced a rough set-based technique to automatically extract configuration rules from historical design data, aiming to reduce reliance on domain expertise while improving customization efficiency. Similarly, (He et al) proposed a data-mining approach that combines PCA and association rule mining to uncover latent relationships between customer specifications and product configurations, supporting adaptable classification.

In parallel, recent developments in e-commerce product classification illustrate how traditional rule-based systems are evolving through machine learning. The study by (Cotacallapa et al) presents a flat-hierarchical hybrid model that combines flat and local classifiers to address the limitations of each. Trained on a four-level product taxonomy, their model showed improved performance over standalone approaches and demonstrates how hybrid classification architectures can improve accuracy in large-scale, multi-level product datasets.

Current research highlights the effectiveness of neural networks in capturing complex, non-linear feature interactions for product family classification. These models support scalable and automated classification, making them particularly suitable for manufacturing environments where product families often exhibit intricate hierarchical structures. However, traditional rule-based approaches remain valuable due to their interpretability and reliance on expert knowledge, allowing for transparent decision-making and easier integration in domains where explainability and domain expertise are critical.

2.3 Clustering methods for Case-Based Reasoning

Case-Based Reasoning (CBR) is a problem-solving approach that solves new problems by reusing solutions from previously encountered, similar cases. In industrial contexts, a major challenge lies in the efficient retrieval of relevant cases from large and complex case bases, where the number of stored records can be substantial and highly diverse. To address this, several research efforts have focused on enhancing CBR retrieval performance, with clustering techniques emerging as a key solution. These methods help reduce search space, improve retrieval speed and increase the relevance of retrieved cases by organizing them into meaningful groups based on shared characteristics set.

Several approaches have been proposed to enhance the efficiency and accuracy of CBR retrieval, especially when handling large, heterogeneous case bases common in industrial applications. Clustering-based methods are frequently employed to reduce search complexity by grouping similar cases, thus limiting comparisons during retrieval. For example, (Mansoul and Atmani) introduced a clustering technique that partitions the case base to improve retrieval efficiency while preserving solution relevance in dynamic environments. Building on this, (Zhang et al) developed the Hierarchical Case-Based Classifier (HCBC), which integrates conceptual clustering within a hierarchical lattice structure. This multi-level organization allows progressive narrowing of the search space, significantly enhancing both classification accuracy and retrieval speed. In a process-oriented domain, (Müller and Bergmann) leveraged process-based clustering that aligns case organization with operational workflows, resulting in improved similarity matching through domain-relevant grouping. Additionally, (Yang and Wu) proposed an ensemble strategy that fuses clustering with decision forests to enhance case relevance estimation, supporting interactive CBR decision-making.

While effective, many of these methods entail substantial computational overhead, require domain-specific parameter tuning, or struggle with efficient nearest-neighbor search in continuous, moderate-dimensional numeric spaces. This is where KD-Trees remain a justified and practical choice. As a spatial indexing technique, KD-Trees partition feature space based on numerical values, enabling logarithmic-time nearest-neighbor queries for low- to moderate-dimensional data. Importantly, (Sample et al) addressed classical KD-Tree limitations by introducing an optimized search algorithm that combines Depth-First Branch and Bound (DFBB) with intelligent node ordering and pruning heuristics.

This work demonstrated that although KD-Trees inherently face challenges in high-dimensional spaces, careful traversal optimization can reduce search times by up to 80% compared to naïve methods. This optimization makes KD-Trees highly suitable for manufacturing datasets typical of industrial CBR systems, where feature vectors (often 8 to 15 dimensions) are standardized and cases pre-clustered by product family.

These studies highlighted the effectiveness of clustering techniques to streamline the retrieval process. By leveraging clustering to reduce the search space, the system can make more accurate and efficient decisions. Building on this, the KD-Tree method is particularly well-suited to our dataset’s numerical and geometric nature, offering a scalable, deterministic, and computationally efficient retrieval approach. Its natural compatibility with continuous numeric features in our manufacturing time estimation.

3 Research question

Although Case-Based Reasoning (CBR) and Machine Learning (ML) have demonstrated improvements in manufacturing time estimation accuracy, most existing approaches remain limited in scope. They often depend on static datasets, require extensive manual feature engineering, and are typically tailored to a single product family. Critically, these methods lack the flexibility to adapt to the high variability and complexity inherent in ETO environments—where product families can differ significantly in terms of geometry, materials and production constraints. This limits their generalizability and reduces their effectiveness in real-world industrial applications.

A major gap in the literature is the absence of an automated framework capable of performing manufacturing time estimation across multiple, heterogeneous product families. Current solutions rarely tackle key challenges such as automatic classification of new products and dynamic data preprocessing (including transformation, imputation and outlier mitigation) and scalable retrieval methods suited for high-dimensional case spaces. Consequently, there remains a lack of practical, data-driven tools that can deliver accurate and adaptive time estimation tailored to the complex of real-world ETO manufacturing.

To address these gaps, this research introduces an automated, data-driven framework for manufacturing time estimation across multiple product families. The proposed solution employs a hybrid learning architecture that integrates product family classification with case-based reasoning. Central to the framework is the use of either Rule-Based Learning (RBL) or Artificial Neural Networks (ANN) for automated product family assignment, combined with feature preprocessing techniques to manage data variability. To enhance retrieval efficiency, a KD-Tree-based mechanism is used to support Case-Based Reasoning (CBR), enabling precise and computationally efficient similarity searches. Designed to accommodate the structural diversity typical of ETO environments, this end-to-end system has been validated through an industrial case study. The results demonstrate substantial improvements in prediction accuracy—evaluated using MAE and RMSE—as well as significant gains in computational efficiency when estimating manufacturing time across complex, multi-family production settings.

4 Research methodology

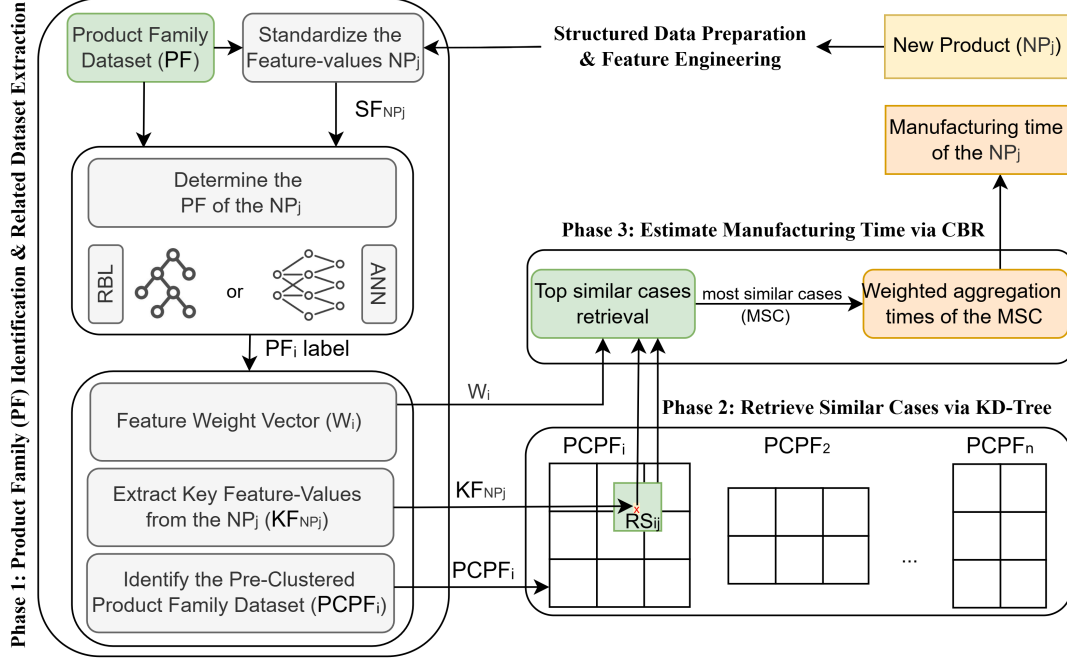


FIG. 1 – Integrative Machine Learning Framework for Manufacturing Time Estimation

The proposed integrative framework for a manufacturing time estimation approach is illustrated in Fig.1. It begins with the description of the new product (NP_j) for which we want to estimate the manufacturing time. The new product will first be assigned to one of the predefined product families within the Product Family Dataset (PF), ensuring that subsequent comparisons and retrievals are made within the most relevant subset of historical data. NP_j is characterized by a set of features such as geometric dimensions, material properties, and manufacturing parameters. To ensure consistency with historical records, the feature vector is automatically standardized through a data preprocessing pipeline, which includes automated handling of missing values, feature scaling, and format alignment. Once standardized, the new product undergoes a classification process—using either a Rule-Based Learning (RBL) system or an Artificial Neural Network (ANN)—to determine its corresponding product family (PF_i). This classification step is critical, as each product family embodies distinct manufacturing behaviors that directly impact manufacturing time. Based on the assigned PF_i , the system retrieves the associated Pre-Clustered Product Family Dataset ($PCPF_i$)—a filtered subset of the historical dataset containing only observations that belong to the same product family. Alongside this, it also retrieves the corresponding Feature Weight Vector (W_i), which represents the relative importance of each feature, determined through prior significance analysis specific to that family. The new product is then projected into the family-specific feature space by extracting its Key Feature-Values (KP_{NP_j}), meaning only the subset of feature values that align with the feature set are used within PF_i . This tailored representation forms the foundation for the subsequent similarity-based case retrieval process.

Following this initial classification, the framework focuses on retrieving the most relevant historical cases. Leveraging the pre-clustered dataset ($PCPF_i$), a KD-Tree is constructed to organize historical instances into a hierarchical structure that enables efficient nearest-neighbor search. The extracted feature vector (KP_{NP_j}) of the new product is used as a query input to the KD-Tree, which computes distances between this vector and existing points in the pre-clustered feature space. Based on these distance calculations, the KD-Tree efficiently identifies and retrieves the top- k historical cases that are most similar to the new product. This subset of cases forms the Relevant Case Set (RS_{ij}), serving as the foundation for subsequent estimation using CBR. By narrowing the search space to only pertinent examples, the system not only accelerates retrieval but also ensures that subsequent estimations are grounded in closely matched manufacturing histories.

With the RS_{ij} retrieved, the final phase estimates the manufacturing time of NP_j through a Case-Based Reasoning (CBR) approach. The framework first computes Local Similarity scores by comparing individual feature values between the new product and each case in RS_{ij} . These are then aggregated—using predefined feature weights—to calculate a Global Similarity score for each case. Based on these scores, the cases are ranked, and the top-most similar instances are selected for the final estimation step. Manufacturing time is then estimated by weighting each retrieved case’s actual manufacturing time according to its normalized similarity score, ensuring that more relevant cases exert greater influence. This weighted aggregation strategy increases robustness against variability, producing a more stable and accurate prediction. Altogether, the integrated process—from family classification, to efficient retrieval, to similarity-weighted reasoning—offers a scalable, interpretable and highly effective solution for manufacturing time estimation in complex, multi-family production environments.

To ensure that this integrative framework operates effectively, the following section focuses on structured data preparation and feature engineering. Before deploying any learning or reasoning algorithms, the data must be cleaned, transformed and aligned with the operational semantics of each product family. These preparatory steps are critical for minimizing noise, handling missing or qualitative inputs.

4.1 Structured Data Preparation and Feature Engineering

In real-world manufacturing environments, data is often noisy, incomplete and heterogeneous. Products may differ substantially in geometry, material, production constraints and process complexity. Treating all product instances as part of a single, undifferentiated dataset imposes significant challenges for predictive modeling. It introduces noise and dilutes the relevance of patterns, making it harder for machine learning algorithms to generalize effectively.

The objective of the data preparation phase is to transform raw operational records into a structured format suitable for modeling and analysis. Each row must correspond to a unique product instance, while columns represent manufacturing-relevant attributes such as diameter, production quantity and yield strength. This structured dataset forms the foundation for downstream tasks.

The usual data preparation workflow includes the following steps :

- **Target Validation** : To ensure the reliability of supervised learning, all instances missing the target variable—manufacturing time in this case—are removed. Imputing targets risks training the model on fabricated outcomes, leading to misleading patterns and compromised evaluation metrics.

This step maintains the integrity of the training process by ensuring that each observation includes both input features and a corresponding output for model learning and evaluation.

- **Feature Selection** : To mitigate dimensionality issues and improve model interpretability, we apply both correlation-based and statistical-based feature selection to identify the most relevant features for each PF. This is done using the Spearman correlation to capture monotonic relationships between numerical features and the target variable. Unlike Pearson correlation, which measures only linear associations, Spearman is non-parametric to outliers and skewed distributions, which is an important consideration given the non-linear nature and heterogeneity of manufacturing data.
- **Missing Value & Outlier Handling** : To maintain data quality and reduce bias, rigorous preprocessing is performed to address missing values and detect outliers. Median imputation is used for numerical features due to its robustness against skewed distributions and its ability to preserve the central tendency without being affected by extreme values unlike mean imputation. We observed that median imputation consistently led to lower error rates. For handle outlier points, from Singh and Upadhyaya (2012) multiple techniques have been assessed, including : IQR Inter-quartile Range : statistical method based on percentile thresholds. This is simple and effective for identifying mild to moderate outliers in unimodal distributions. Isolation Forest : an unsupervised machine learning algorithm that is particularly effective for high-dimensional datasets. It isolates outliers by random partitioning, making it efficient and scalable. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) : a clustering-based approach that identifies outliers as low-density points, used in datasets where outliers form natural clusters or noise. After empirical testing across product families, IQR-based filtering was selected as the most appropriate method. It consistently achieved lower prediction errors while maintaining the natural distribution of key features, ensuring the cleaned dataset remained representative and suitable for learning. More advanced methods such as Isolation Forest and DBSCAN either removed too many instances or distorted the data geometry, which affected downstream performance.
- **Feature Transformation** : To enrich the dataset and preserve the significance of qualitative variables, categorical features—such as material types—are transformed into meaningful quantitative descriptors. This is achieved by integrating external domain knowledge or engineering databases (e.g., mapping material types to their mechanical properties such as yield strength), enabling the model to better capture the impact of those attributes on manufacturing time.

4.2 Product Family Identification and Related Dataset Extraction

In this phase, our objective is to enhance estimation accuracy by ensuring that each product is analyzed within the context of its specific product family (PF), rather than across a generalized dataset. This is especially crucial in our case, where the number of product families is large and each family is characterized by its own distinctive set of features. Classifying new instances product data into the correct PF allows downstream tasks—such as similarity retrieval and time estimation — to operate on relevant subsets.

By narrowing the case base to a relevant product family, we ensure that similarity retrieval is performed in a meaningful context, ultimately improving predictive performance and interpretability. This segmentation not only enhances predictive performance and interpretability but also significantly reduces computational overhead by limiting the search space, thereby accelerating subsequent processing steps.

The classification process begins by transforming the new product’s input features into a standardized format. This includes selecting features that are common across all families and applying scaling techniques to ensure uniformity. Once standardized, the input is passed through one of two mutually exclusive classification models, depending on the data availability and complexity of the problem : Rule-Based Learning (RBL) or Artificial Neural Networks (ANN)

The chosen model outputs a product family label, which is then used to extract the relevant pre-clustered product family dataset. Once standardized, the input is passed through one of two mutually exclusive classification models—RBL or ANN—depending on data characteristics and model performance. Through experimental evaluation across multiple product families, we observed that RBL consistently outperformed ANN in validation accuracy for the majority of cases, particularly where domain knowledge allowed for well-defined threshold rules. Therefore, RBL is used as the primary classifier. However, in cases where RBL rules may be less precise or ambiguous—especially for families with overlapping feature boundaries—ANN is used as a complementary model. In such cases, the ANN serves as a validation layer to double-check predictions or to act as the primary model if it demonstrates superior performance (e.g., in Product Family OT). This hybrid strategy allows the system to balance interpretability and accuracy, adapting flexibly to the variability across different product families.

Further details of the two classification approaches are outlined below :

- **Classification of Artificial Neural Networks (ANNs)** : ANNs are supervised machine learning models inspired by the architecture of the human brain. They comprise layers of interconnected neurons, typically including an input layer, multiple hidden layers, and an output layer. Each neuron processes a weighted sum of its inputs, applies a non-linear activation function, and propagates the signal forward. This architecture enables the network to model complex, non-linear relationships between input features and target labels.

In our context, Artificial Neural Networks (ANNs) have proven particularly effective for product family classification because the decision boundaries between families are often non-linear and overlapping. This is especially true for families such as T and OT, where feature distributions intersect and simple rule-based models struggle to separate them reliably. Features such as geometric dimensions, material characteristics, and operational constraints interact in complex ways. Neural networks are capable of capturing these intricate patterns and generalizing from large, high-dimensional, and potentially noisy labeled datasets.

We employ a feedforward neural network architecture for multi-class classification, where each output node corresponds to a specific product family. The model is trained to minimize the categorical cross-entropy loss, using softmax activation in the output layer to produce class probabilities. The pipeline begins with data aggregation, where all labeled product families are merged into a unified dataset. Key features common across all families are selected and standardized to form a consistent input space. The dataset is split into training, validation and test sets (70% / 20% / 10%) to ensure generalizability and avoid overfitting.

The model consists of an input layer matching the number of selected features, three hidden layers with ReLU activation, and dropout layers to mitigate overfitting. Training is conducted using the Adam optimizer and backpropagation. Callbacks such as early stopping and model checkpoints are employed to monitor validation performance and preserve the best-performing model. The final model is evaluated on the test set using metrics such as accuracy, precision, recall and F1-score and then saved for future inference.

- **Rule-Based Learning (RBL) Classification** : This is a symbolic, interpretable approach that classifies data by applying a set of predefined "if-then" rules. These rules are based on logical conditions involving feature values, statistical thresholds and domain-specific knowledge. RBL is particularly advantageous in manufacturing contexts, where transparency and expert-driven criteria are crucial.

In our context, RBL offers several advantages, including that : knowledge from manufacturing experts—such as allowable tolerances, geometrical ranges, or process constraints, can be encoded directly into the rule system.

By collaborating with manufacturing engineers to gather domain knowledge about key features per product family (e.g., geometric limits, tolerances, typical material properties), the logical rules are defined using feature thresholds and value ranges. The rule set is implemented using conditional logic (Python’s ‘if-elif’ blocks). Each instance is passed through the rules and match determines the classification. The predicted labels are compared against known labels from a validation set, and misclassifications are analyzed to refine existing rules or introduce new ones, ensuring continuous improvement.

By accurately classifying products into their respective families, both ANNs and RBL enable downstream tasks to function within narrowed, contextually relevant case bases. The resulting product family label also serves as a key to extract the associated pre-clustered subset of data, which forms the input for all subsequent modeling phases.

4.3 Retrieve Similar Cases via K-Dimensional Tree

Similarity computation is a core step in Case-Based Reasoning (CBR), where a new product is compared to historical cases based on feature distances to identify the most relevant matches. However, in our context, this process becomes increasingly burdensome as the size of the historical dataset grows. For example, each product family PF_i , we often manage thousands of instances—such as over 2676 records for family “J” alone. Performing similarity calculations across all entries not only incurs high computational cost and latency but also increases the likelihood of including marginally relevant or noisy cases. Therefore, it is essential to implement an efficient retrieval mechanism that can narrow the search space without compromising the quality of the results.

To address this, we propose the K-Dimensional Tree (KD-tree) - a spatial indexing structure designed for efficient nearest-neighbor search in multidimensional datasets - to structure and partition the dataset for each product family. When a new product specification is introduced, its key feature values are transformed into a feature vector that serves as a query to the KD-Tree. The KD-Tree algorithm traverses its hierarchical structure using recursive partitioning to identify the region of the feature space where the new product is likely to reside, allowing us to retrieve the similar instances.

This process significantly reduces the search space, optimizes computation time and improves retrieval relevance. By focusing on a contextually close subset—denoted as the Relevant Similar Case Set (RS_{ij})—we enhance both the efficiency and the quality of the subsequent Case-Based Reasoning (CBR) step. This refined set contains only those cases that are most comparable to the new product, ensuring that downstream similarity assessment and time estimation are based on truly relevant historical data. The case retrieval process using KD-Tree operates on the pre-clustered product family subset ($PCPF_i$) identified during the preprocessing phase. In this earlier step, the general dataset was segmented into smaller, $PCPF_i$, each containing only observations relevant to that family. Within each subset, clustering techniques were applied to organize the data for efficient retrieval. Using the fixed set of relevant features for the assigned product family, the new product’s feature values are used to guide the retrieval. The process involves the following steps :

- **Dataset Structuring** : The historical dataset is recursively partitioned into a hierarchical tree structure. In this structure, the data is repeatedly divided into smaller parts based on the most important features (such as diameter, thickness or quantity). Each point of division is called a “node” and each node decides how to split the data depending on one feature at a time. This creates a tree-like layout, where similar cases are grouped closer together. This structure allows the system to quickly narrow down the search to the most relevant portion of the data, avoiding the need to compare the new product against every single historical case.
- **Query Positioning** : The new product’s standardized feature vector is compared against the tree structure to identify its approximate position. This efficiently narrows down the search to the most relevant region of the dataset.
- **Nearest Neighbor Retrieval** : A distance-based search is performed within the identified region to retrieve the most similar historical cases. These nearest neighbors form the relevant case subset RS_{ij} , which is then passed to the CBR module for final time estimation.

This approach significantly improves the speed of case retrieval, particularly in low- to moderate-dimensional datasets—typically those with 10 to 15 features or fewer—where the KD-Tree provides near-logarithmic search efficiency, reducing runtime from linear ($O(N)$) to approximately $O(d \cdot \log_2 N)$.

4.4 Estimate Manufacturing Time via Case-Based Reasoning

Case-Based Reasoning (CBR) is a memory-based learning approach that generates predictions for new instances by adapting solutions from previously solved cases exhibiting similar characteristics. In the proposed framework, after identifying the most relevant subset of historical data via KD-Tree-based retrieval, CBR is applied to estimate the manufacturing time by systematically evaluating the similarity between the new query and the retrieved cases. As illustrated in Fig. 2, the CBR process in our system operates through three key steps. First, retrieval of the most similar historical cases are retrieved. The manufacturing time estimation step then combines both reuse and revision logic. The retrieved cases are ranked based on similarity scores, and a weighted aggregation of their manufacturing times—using normalized similarity weights—is performed to compute the final estimate. Finally, the new validated case is saved in the historical database.

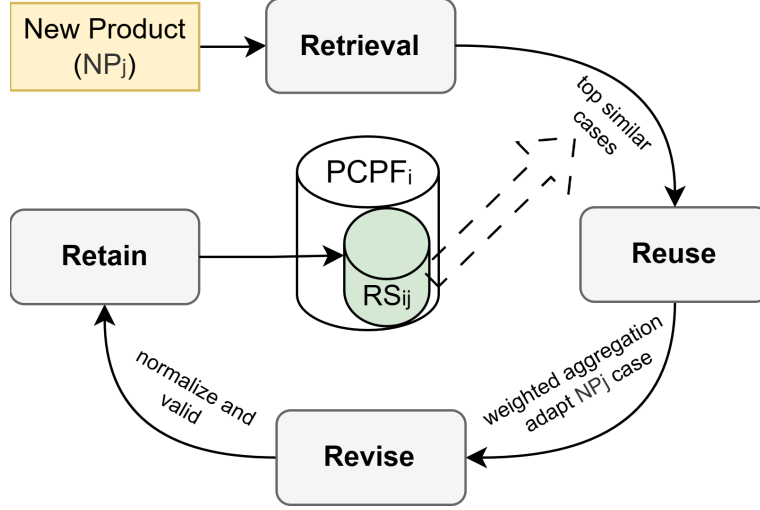


FIG. 2 – CBR Process in Manufacturing Time Estimation Framework

At the core of the reuse phase lies the computation of similarity, which determines how much influence each retrieved case should have on the final estimate. This involves two levels of analysis : **Local Similarity (LS)**, which assesses how closely each individual feature of a historical case matches the new product ; and **Global Similarity (GS)**, which synthesizes these comparisons into an overall similarity score using predefined feature weights.

Local Similarity (LS) : Local similarity measures the closeness between individual feature values of the new product and those of a historical case. In this framework, we restrict the similarity computation to numerical (cardinal) features, as these allow for meaningful distance-based comparison. These local similarities serve as the basis for the global similarity calculation that follows. :

$$LS(f_j) = 1 - \frac{|f_j^{\text{new}} - f_j^{c_i}|}{\max(f_j) - \min(f_j)} \quad (1)$$

where f_j^{new} is the value of the j^{th} feature for the new case, and $f_j^{c_i}$ is the corresponding value in the i^{th} historical case. This formula ensures all local similarities are normalized within the $[0, 1]$ range.

Global Similarity (GS) : Global similarity is computed as a weighted sum of all local similarities, with the weights reflecting the relative importance of each feature :

$$GS(c_i) = \sum_{j=1}^m w_j \cdot LS(f_j) \quad (2)$$

Here, w_j is the weight assigned to the j^{th} feature and m is the total number of features.

Weighting Strategy : in CBR, assigning the appropriate weight to features is critical, as not all attributes contribute equally to predicting manufacturing time. Proper weighting ensures that more influential features have a greater impact during a similarity computation, which directly affects the accuracy of the time estimation.

Various strategies exist for determining feature weights, including expert judgment, Analytic Hierarchy Process (AHP), and automated learning-based methods such as neural networks. In this study, feature weights w_j were obtained through a data-driven statistical approach for its transparency, scalability and reduced dependency on subjective input.

Our process involves two main steps : First, we apply Principal Component Analysis (PCA) to assess the significance of each feature with respect to variance in the dataset. PCA helps identify which features contribute most to distinguishing between different cases. Second, we compute correlation coefficients (Spearman) between each feature and the target variable (manufacturing time). These correlations provide a direct measure of how strongly each feature is associated with the outcome. By combining the insights from PCA and correlation analysis, we assign higher weights to features that are both statistically significant and positively correlated with the prediction target. This strategy allows us to enhance retrieval precision while keeping the weighting mechanism interpretable and computationally efficient. The final weight set is selected based on its ability to minimize Mean Absolute Error (MAE) during validation.

Manufacturing Time Estimation : To enhance prediction robustness and accuracy, the final manufacturing time estimate T_{mt} is not derived from a single nearest case, but from a weighted aggregation of the top n most similar historical cases. This approach accounts for multiple relevant examples, rather than relying solely on one potentially atypical instance, thereby reducing sensitivity to outliers and noise in the case base. During our experiments, we observed that aggregating the top- n similar cases using similarity-based weights consistently resulted in lower Mean Absolute Error (MAE) compared to using only the single most similar case or unweighted averaging.

The normalized weight of each case is calculated using its global similarity GS_i , and the final estimated manufacturing time is given by Equation 3 ensures that more similar cases have a proportionally higher influence on the prediction, enhancing the contextual relevance and robustness of the estimate.

$$T_{mt} = \sum_{i=1}^n \alpha_i \cdot T_i, \quad \text{where} \quad \alpha_i = \frac{GS_i}{\sum_{j=1}^n GS_j} \quad (3)$$

5 Developments and results

5.1 Structured Data Preparation and Feature Engineering

This study applies the proposed approach to a practical case involving a small metallurgy company in France that produces customized products according to client-specific demands. This company manufactures various product families, each characterized by geometric dimensions, material types and specific manufacturing processes.

The initial dataset (Table 1) encompasses 19 Product Families (PFs), each containing a variety of manufacturing-related attributes. The primary predictive target was the manufacturing time (TP). Due to a high occurrence of missing values, particularly in the target column, an initial cleaning step involved removing instances with undefined TP values to maintain the integrity of supervised learning tasks.

Following this, a feature selection process was applied using the Spearman correlation analysis to identify the variables that are most strongly associated with the manufacturing time for each product family. This step significantly reduced dimensionality while preserving its predictive abilities, as presented in Table 2.

TAB. 1 – Initial dataset

Code	Product Family	Expanded Length	Amorce	Quantity	Angle	Bending Diameter	Thickness	Height	Circle Diameter	Designation	Description	Material	Material Class	...	Manufacturing Time
21865	J	1533	367	1	152	488	20		2000	Fourniture matière ...	secteur(s) cercle à 152.6 ...		acier1	...	8.5
21866	J	1460	0	60	64.3	1104.3	2.9		48.3	Fourniture matière ...	secteur(s) cercle à 66 ...		acier1	...	30
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
21857	20	4047	500	1	175	2645	5	1046	Ø INT 2640	Roulage à façon ...	secteur virole 175 ...	S235JR	acier1	...	1.25
21841	20	1351	0	6	360	430	10	2000	Ø INT 420	Fourniture matière et ...	virole Ø ext 440 ...	P355GH	acier1	...	1.8
21847	B	1600	500	1	92.22	1988	20	80	R int 954	Fourniture matière et ...	secteur cercle en plat ...	S235JR	acier1	...	0.95
21848	B	1700	500	1	97.69	1994	20	80	R int 1037	Fourniture matière et ...	secteur cercle en plat ...	S235JR	acier1	...	0.95

TAB. 2 – Feature Selection

Product Family	Expanded Length	Amorce	Quantity	Angle	Bending Diameter	Thickness	Height	Circle Diameter	Material	Manufacturing Time
J	1533	367	1	152	488	20		2000		8.5
J	1460	0	60	64.3	1104.3	2.9		48.3		30
...	...	...	...	...	...	...	...	...	...	...
20	4047	500	1	175	2645	5	1046		S235JR	1.25
20	1351	0	6	360	430	10	2000		P355GH	1.8
B	1600	500	1	92.22	1988	20	80		S235JR	0.95
B	1700	500	1	97.69	1994	20	80		S235JR	0.95

Next, we addressed data quality issues through missing value imputation and outlier handling. Median imputation was chosen for its robustness against skewed distributions. To detect outliers, we evaluated several techniques, including ISO and DBSCAN, but found that the IQR method performed best on our dataset. The cleaned version of the dataset is illustrated in Table 3, where missing and undefined values were standardized using a placeholder (-999). This ensured clear interpretability during model training and inference, especially when certain features not relevant to specific PFs.

TAB. 3 – Missing Value and Outlier Handling

Product Family	Expanded Length	Amorce	Quantity	Angle	Bending Diameter	Thickness	Height	Circle Diameter	Material	Manufacturing Time
J	1533	367	1	152	488	20	-999	2000	INOX304	8.5
J	1460	0	60	64.3	1104.3	2.9	-999	48.3	INOX304	30
...	...	...	...	...	...	...	...	...	...	...
20	4047	500	1	175	2645	5	1046	-999	S235JR	1.25
20	1351	0	6	360	430	10	2000	-999	P355GH	1.8
B	1600	500	1	92.22	1988	20	80	-999	S235JR	0.95
B	1700	500	1	97.69	1994	20	80	-999	S235JR	0.95

To enrich categorical material descriptors and transform them into machine-readable features, we performed a feature transformation. Specifically, we mapped each material type to its corresponding mechanical property (i.e., yield strength) sourced from the SteelNumber engineering materials database via automated retrieval. The outcome of this transformation step is shown in Table 4.

TAB. 4 – Feature Transformation

Product Family	Expanded Length	Amorce	Quantity	Angle	Bending Diameter	Thickness	Height	Circle Diameter	Material	Manufacturing Time
J	1533	367	1	152	488	20	-999	2000	215	8.5
J	1460	0	60	64.3	1104.3	2.9	-999	48.3	215	30
...	...	...	...	...	...	...	...	...	...	...
20	4047	500	1	175	2645	5	1046	-999	235	1.25
20	1351	0	6	360	430	10	2000	-999	265	1.8
B	1600	500	1	92.22	1988	20	80	-999	235	0.95
B	1700	500	1	97.69	1994	20	80	-999	235	0.95

Following these operations, a consolidated and ready-to-use dataset was obtained, as shown in Table 5. This version features clean, structured, and numerically enriched inputs compatible to the proposed hybrid learning chain (HMLC). Each row represents a unique product instance, with columns reflecting key manufacturing features such as : diameter, yield, quantity and others.

TAB. 5 – Processed dataset ready to be used by the HMLC

Product Family	Expanded Length	Amorce	Quantity	Angle	Bending Diameter	Thickness	Height	Circle Diameter	Yield	Manufacturing Time
G	4970	500	1	360	1582	3.5	35	-999	235	1.22
G	4086	600	1	120	3902	10	100	-999	235	4.09
G	1947	500	1	360	620	3	20	-999	210	1.09
G	3691	500	2	180	2350	5	50	-999	210	2.25
20	1533	367	1	360	488	20	2000	-999	215	8.5
20	6873	0	1	360	2188	12	2430	-999	245	6.75
20	1266	0	1	360	403	3	3000	-999	245	2
20	748	458	2	48	1788	12	686	-999	275	2.4
B	1700	500	1	97.70	1944	20	80	-999	235	0.95
B	1831	200	1	328.87	638	15	70	-999	210	1.59
B	1065	0	3	60.87	2005	20	125	-999	210	3
B	4536	500	1	371.87	1400	20	60	-999	345	2.25
B	2246	200	1	302.79	850	15	150	-999	345	5.32
B	3786	200	1	405.46	1070	15	150	-999	245	4
B	3996	0	1	380.95	1202	10	80	-999	245	1.75
OT	5668	800	2	253.32	2564	8	100	-999	235	3.15
OT	5315	500	1	359.96	1692	5	60	-999	235	1.5
OT	2652	500	2	181.32	1676	-999	10	-999	690	3.5
OT	4050	600	2	389	1213	3	60	-999	210	2.34
OT	4916	500	1	355.45	1585	5	60	-999	235	2.1
OT	4995	600	5	359.99	1590	5	60	-999	235	6
...	...	...	...	...	...	...	...	...	...	...
J	2111	1200	1	179.99	1344	4	-999	168.3	235	7.5
J	2500	0	3	135.38	2116.1	2	-999	76.1	290	2.59
J	4398	500	3	119.99	4200	4	-999	60.3	215	4
J	4272	500	2	359.95	1360	2	-999	26.9	235	4
J	3355	500	1	179.98	2136.1	2	-999	76.1	210	1.49
J	516	402	3	159.81	370	2.77	-999	42.4	290	5.29
J	5400	600	2	120.39	5140	2	-999	139.7	210	3.44
J	621	479	2	184.83	385	3.91	-999	60.3	205	4.75

Additionally, the initial set of product families (PFs) was provided by the company based on internal classification standards. To enhance the robustness of the model and ensure its statistical validity, we refined this structure through a rule-based learning approach that was supported by expert recommendations. By analyzing value ranges and the availability of key manufacturing features, products that exhibit similar behavior were systematically grouped. Product families with more than 500 instances and consistent feature patterns were retained as standalone categories. In contrast, under-represented families were consolidated into an auxiliary group labeled “OT” (Other), following expert guidance, to maintain model generalizability without compromising data integrity.

Informed by expert insights and validated through Principal Component Analysis (PCA), material properties have been identified as a key determinant in predicting manufacturing time. As shown in Fig.3, the PCA results reveal that several material-related variables—such as different material classes (acier1, inox, aluminium, etc.)—exert significant influence on the principal components explaining the variance in manufacturing time. As material type was initially a categorical variable, it was transformed into a numerical format by incorporating relevant mechanical properties, notably yield strength. These values were automatically retrieved through web scraping from a dedicated engineering materials database, SteelNumber, which provides comprehensive mechanical and chemical properties for industrial-grade materials. This transformation not only enriched the feature space but also contributed to improved accuracy in the model estimation, as is presented later in Table 7 in section 6.

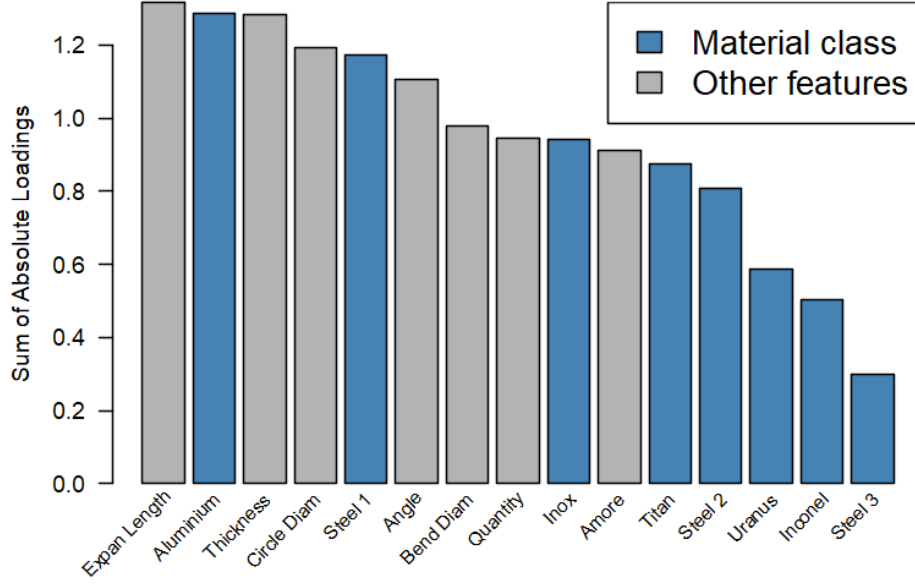


FIG. 3 – Feature Importance from PCA

5.2 Product Family Identification and Related Dataset Extraction

In our industrial use case both classification approaches RBL and NN were applied to assign products to one of eight predefined product families. The classification model was trained on a standardized and merged dataset created through prior feature analysis. Features with the greatest predictive impact were selected to ensure classification. This classification step ensures that each product is associated with the most relevant family, allowing subsequent steps like case retrieval and similarity assessment to focus only on family-specific data.

5.2.1 Neural Network for the Classification of Product Families

The Neural Network model was trained on a standardized dataset containing 12 key features, selected via prior feature analysis and correlation filtering. The architecture consisted of a dense layer with 64 ReLU-activated units and a softmax output layer. Class weights were computed to address label imbalance, the model was trained with a batch size of 128 and a learning rate of $1e-5$. Final model checkpoints and performance logs were archived for downstream use in CBR.

As summarized in Table 6. On the validation set, the model achieved an accuracy of 96.32%, with a corresponding F1-score of 96.28%, indicating a balanced trade-off between precision and recall. The slightly higher performance on the validation set compared to training and test sets (Train : 95.98%, Test : 96.05%) suggests good generalization capability without overfitting. Precision and recall values remain above 96% across all sets, reflecting the model's robustness in correctly identifying product family labels, even in the presence of class imbalance.

TAB. 6 – Neural Network Evaluation

Set	Num of Obs	Accuracy	F1_score	Precision	Recall
Train	4402	0.9598	0.9590	0.9622	0.9598
Test	1267	0.9605	0.9593	0.9619	0.9605
Valid	625	0.9632	0.9628	0.9677	0.9632

5.2.2 Rule-Based Learning for the Classification of Product Families

The RBL strategy was implemented through the following stages :

- **Feature Selection and Analysis** : This stage began by identifying the most relevant features that influence manufacturing time. This involves examining correlation matrices to assess linear relationships and analyzing feature distributions to understand variability and outlier sensitivity.
- **Threshold and Range-Based Rule Creation** : Once the key features are selected, we examine their value ranges and distributions to determine thresholds or combinations that distinguish product families. Eg : *If diameter > 4055 and quantity < 7, then Family M*). A decision tree model was also trained to extract interpretable splitting rules for ambiguous families like “G” and “M”.
- **Rule Formulation and Validation** : Using the derived patterns, we construct a set of classification rules—if-then conditions—that allocate each product to a product family. These rules are tested on a standardized and labeled dataset to ensure generalizability and accuracy.

The implementation of this phase yielded promising results which will be discussed in more detail in Section 6. The classifier successfully segmented the dataset into distinct product families with high fidelity, leading to a more focused case retrieval and overall improved model accuracy. However, some limitations remain. The manual inspection of rule logic still requires domain expertise to refine edge cases. Furthermore, classification errors—although rare—can propagate into the CBR phase, affecting final predictions. Nonetheless, the benefits outweigh the drawbacks, as family-specific modeling consistently demonstrated a lower MAE compared to models trained on a mixed dataset.

5.3 Retrieve Similar Cases via K-Dimensional Tree

In our practical deployment, the KD-Tree method was employed to optimize the retrieval efficiency of relevant manufacturing cases before estimating the production time. Rather than performing exhaustive similarity comparisons across the entire historical dataset, which is computationally intensive, the KD-Tree enables efficient nearest-neighbor search by hierarchically partitioning the feature space. For each new product instance, it uses the corresponding Pre-Clustered Product Family Dataset. A KD-Tree is then constructed using only the key features relevant to that family.

Directly comparing the new instance to all records would be time-consuming. Instead, by querying the KD-Tree built for each family the system rapidly locates a small neighborhood of similar products—e.g., the 100 closest cases in feature space—using logarithmic search complexity. This focused retrieval significantly reduces computational cost while preserving estimation accuracy, making it particularly effective in high-dimensional and large-volume manufacturing datasets.

For instance, in a subset with $N = 300$ and $d = 15$, the brute-force method would require 300 operations, while KD-Tree retrieval would reduce this to approximately $15 \cdot \log_2(300) \approx 15 \cdot 8.2 = 123$ operations—less than half the original effort. Overall, this approach significantly enhances retrieval speed in low- to moderate-dimensional settings (10–15 features).

The KD-Tree not only improved computational speed but also enhanced estimation quality by ensuring that only contextually relevant cases were selected for comparison. This focused retrieval mechanism directly contributed to lower MAE and RMSE scores, as detailed in Section 6.

5.4 Estimate Manufacturing Time via Case-Based Reasoning

In our industrial application, the CBR method was integrated with the KD-Tree retrieval system to estimate manufacturing times. For each new product, the top $k = 5$ most similar historical cases were identified based on selected manufacturing features.

Key components of our implementation included :

- **Min–max normalization** of each feature based on its distribution within each product family, ensuring consistent similarity computation across diverse families.
- **Feature weighting** was optimized through regression techniques and cross-validation. Features such as *yield strength*, *diameter*, *quantity* and others were found to be influential.
- **Similarity thresholding** was applied by discarding cases with $GS < 0.8$, improving the relevance of result.

Experimental results revealed that using the average of the top $n = 5$ most similar cases consistently improved prediction performance. This hybrid approach reduced both MAE and RMSE, validating its suitability for dynamic and variable-rich ETO manufacturing environments.

6 Discussion

Using Mean Absolute Error (MAE) as a key metric, we compare our results against a previous study in the French metallurgy industry (Chehade et al, 2024), which employed KNN and CatBoost for case retrieval and adaptation. While the prior study focused on manufacturing time estimation of product families ‘J’, we extend our analysis to eight product families. However, for clarity, only the results for PF

Table 7 presents the Mean Absolute Error (MAE) between actual and estimated manufacturing times under four different scenarios. These scenarios demonstrate how various elements—such as feature weighting, data transformation and retrieval strategies—impact estimation accuracy and overall model performance.

TAB. 7 – Comparison of the strategies by MAE

Product Family	Number of Observations	Only CBR	Feature Sign + Weight + CBR	Feature Sign + Weight + KD-CBR	Enhance Feature Ana + Weight + KD-CBR
J	293	2.56	1.58	1.45	1.42
B	274	2.81	1.88	1.81	1.69
20	419	1.88	1.89	1.92	1.55
13	82	3.86	1.64	1.42	1.47
G	42	3.79	1.35	1.25	1.2
M	32	1.83	1.48	1.4	1.13
T	59	3.75	1.89	1.88	1.94
OT	92	2.41	2.3	2.13	1.84

The final configuration —Enhanced Feature Analysis + Weighting + KD-CBR— achieves the lowest MAE across nearly all product families, underscoring the robustness and scalability of the proposed framework. This consistency is especially evident among product families with large test samples, such as 20 ($n = 419$), J ($n = 293$) and B ($n = 274$), all of which demonstrate substantial and consistent error reductions. For example, the MAE for family J drops significantly from 2.56 in the baseline to 1.42, while family B improves from 2.81 to 1.69. Although family 20 exhibits some variability in earlier configurations, the final model still achieves a notable reduction to 1.55, indicating that some families may respond more to feature-level enhancements than to retrieval strategies alone.

To further understand the evolution of model performance, we delve into a detailed analysis of product family J, which offers a representative case due to its large test set size and clear pattern of improvement :

1. Only CBR This baseline scenario utilizes Case-Based Reasoning (CBR) without any prior classification or data enhancement. New cases are processed directly through the CBR mechanism. The MAE recorded was 2.56 for PF “J”, serving as a reference for evaluating the impact of additional enhancements.

These cases followed the process outlined in Fig. ??, where new instances were first passed through the classifier model for classification into a subset corresponding to each Product Family.

2. Feature Significance + Weighting + CBR In this configuration, we introduced feature importance analysis through Principal Component Analysis (PCA) and linear modeling to identify and emphasize the most influential attributes. These were then weighted accordingly in the CBR retrieval phase, reducing the MAE to 1.58 for Product Family “J”.

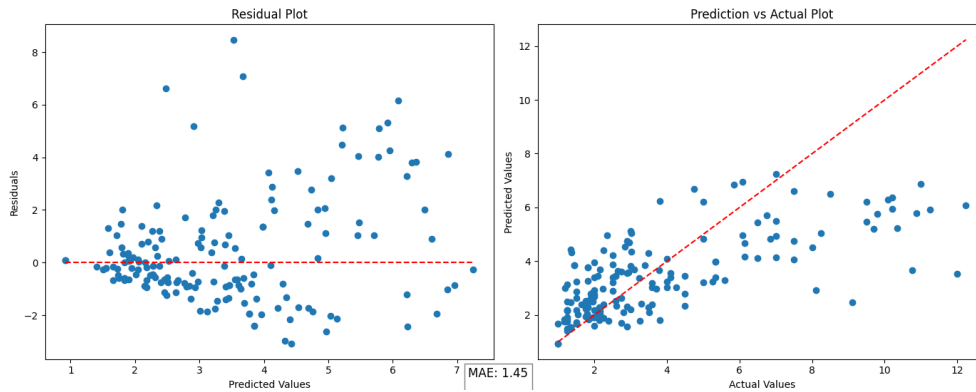


FIG. 4 – Applied KD-Tree

3. Feature Significance + Weighting + KD-Tree + CBR

Building upon Scenario 2, we incorporated KD-Tree-based clustering to optimize to optimize the efficiency and preciseness of case retrieval. This refinement further improved the MAE to 1.45. Residual analysis (see Fig. 4) revealed a heteroscedastic pattern—prediction errors increased with higher target values—indicating robust performance for lower-duration tasks with increasing variance for more complex jobs.

4. Enhanced Feature Analysis (Feature Transform + Feature Significance) + Feature Weight + KD-Tree + CBR

In the most advanced scenario, qualitative attributes—such as material types were transformed into quantitative indicators like yield strength by integrating external domain-specific datasets.

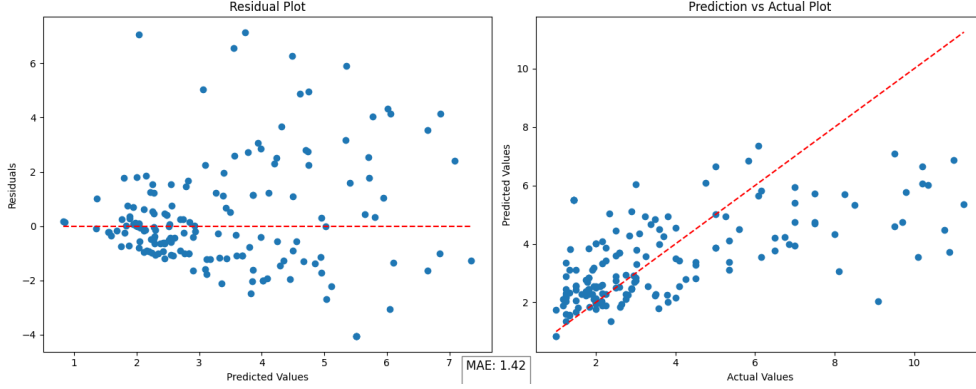


FIG. 5 – Applied Feature Transformation

This enriched representation, combined with structured retrieval, resulted in the best MAE performance of 1.42. As shown in Fig. 5, residual dispersion was slightly reduced compared to Scenario 3, confirming that targeted feature engineering significantly enhances model accuracy.

To contextualize our results, we compared our best-performing model against the CatBoost method employed in a previous study focused on the same Product Family “J” (Chehade et al, 2024). The earlier work achieved a MAE of 1.81 and a RMSE of 3.33 using CatBoost on a test set of 424 samples. In comparison, our proposed method achieved a significantly lower MAE of 1.32 and RMSE of 1.93 on a separate test set of 293 samples. These findings confirm two major points :

First, the reduction in MAE illustrates a more consistent estimation performance, while second, the significant decrease in RMSE demonstrates fewer extreme prediction errors. Together, these improvements highlight the robustness of our model, especially in handling real-world complexity and variability inherent to custom manufacturing processes.

TAB. 8 – Comparison of the strategies by MAE

Method Product Fam	13	20	B	G	J	M	OT	T
Rule-Based Learning	1.4	1.56	1.39	1.2	1.32	1.19	2.26	1.71
Neural Network	1.47	1.55	1.69	1.2	1.42	1.13	1.84	1.94

Table 8 presents the Mean Absolute Error (MAE) between actual and estimated manufacturing times across eight distinct product families, highlighting the effectiveness of our methodology. The results illustrate the impact of improved product classification on estimation accuracy.

Specifically, the table compares MAE values obtained after the classification step, evaluating the performance of Rule-Based Learning (RBL) versus Neural Networks (NN) prior to applying KD-CBR.

One key improvement in our methodology is the enhancement of the classification task. In a previous experiment, classification was handled via a Neural Network (NN) and then we implemented Rule-Based Learning (RBL). As demonstrated in Table 8, RBL achieves a slight improvement in MAE compared to NN, with a reduction from 1.42 to 1.32 for Product Family ‘J’.

Building on this progress, Fig.6 presents the developed graphical user interface (GUI) designed for interactive manufacturing time estimation. The system integrates historical data visualization through control charts and a correlation heatmap, enabling users to observe trends. Upon entering new product specifications, an automated classification module assigns the product to its respective family, ensuring the retrieval of the most relevant subset of historical cases. The case-based reasoning approach is then applied to this subset, providing a time estimation. This interactive tool enhances decision-making by combining real-time user input with data-driven prediction models.

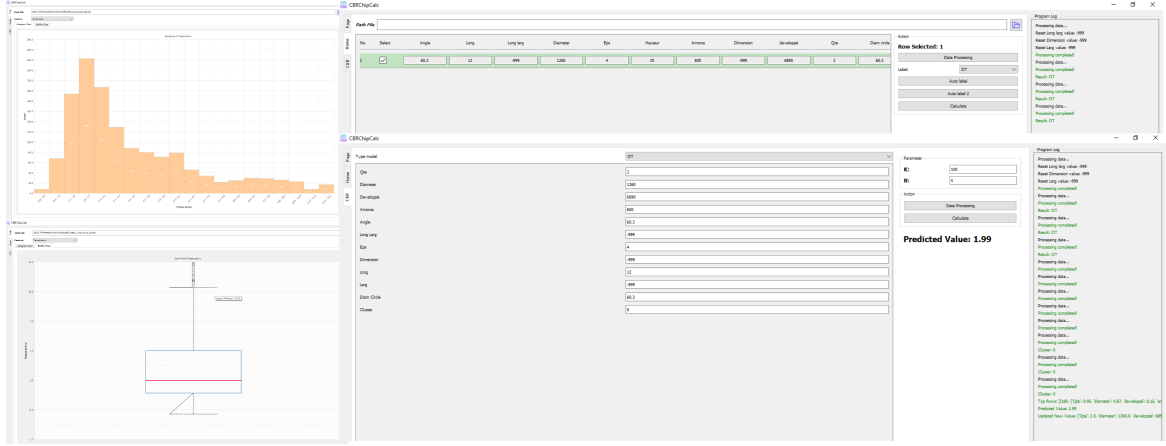


FIG. 6 – Example of GUI for case retrieval and prediction

7 Conclusions

This research has presented an integrative machine learning framework that utilizes neural network (ANN) and rule-based learning (RBL) to enhance product family categorization, thereby improving case-based reasoning (CBR) for manufacturing time estimation. By structuring classification through ANN & RBL, our approach refines the accuracy of case retrieval, leading to more precise predictions. Additionally, experiments aimed at improving data quality—such as enhanced transformation, imputation, and anomaly handling—have strengthened the reliability of input data. To further optimize retrieval efficiency, a KD-Tree algorithm has been incorporated, accelerating case-matching processes and improving scalability. Validated on industrial data from a French metallurgy company, the framework achieves a 10% improvement in Mean Absolute Error (MAE) and reduces retrieval time by 60%, confirming its effectiveness for multi-family, high-variability manufacturing scenarios., delivering reliable manufacturing time predictions across diverse product families in engineer-to-order (ETO) context.

Despite its effectiveness, the framework's current application remains limited to a specific case resulting from confidentiality constraints. Future work will focus on extending the approach to a broader range of product families and refining data processing strategies. Given that inaccuracies and biases in feature extraction may affect predictions, an alternative direction involves leveraging raw input data to minimize preprocessing-related distortions. Additionally, efforts will be directed toward more sophisticated outlier handling and data augmentation techniques to further improve performance. These enhancements will contribute to a more adaptable, scalable solution applicable to diverse manufacturing environments.

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Références

- Aamodt, A., Plaza, E., *Case-based reasoning : Foundational issues, methodological variations, and system approaches*, AI Communications, 7(1), 39–59, 1994.
- Abdelwahed, M.F., Mohamed, A.E., and Saleh, M.A., *Solving the motion planning problem using learning experience through case-based reasoning and machine learning algorithms*, Ain Shams Engineering Journal, 11(1), 133–142, 2020.
- Cateni, S., Colla, V., Vannucci, M., Aramburo, J., and Trevino, A.R., *Outlier detection methods for industrial applications*, Advances in Robotics, Automation and Control, 265–282, 2008.
- Chang, P.C., Fan, C.Y., and Wang, Y.W., *Evolving CBR and data segmentation by SOM for flow time prediction in semiconductor manufacturing factory*, Journal of Intelligent Manufacturing, 20, 421–429, 2009.
- Chehade, M.H. and Sylla, A., *Coupling case-based reasoning and machine learning for manufacturing time estimation*, IFAC-PapersOnLine, 58(19), 941–945, 2024.
- Chehade, M.H., Sylla, A., Diallo, A.R., and Doremus, Y., *Manufacturing time estimation for offer pricing : A machine learning application in a French metallurgy industry*, Engineering Applications of Artificial Intelligence, 109089, 137, 2024.
- Dattani, I. and Magaldi, R.V., *Case-based reasoning and machine learning within the aviation domain*, IASTED International Conference in Artificial Intelligence and Soft Computing, 415–419, 1998.
- Guo, Y., Zhang, B., Sun, Y., Jiang, K., and Wu, K., *Machine learning-based feature selection and knowledge reasoning for CBR system under big data*, Pattern Recognition, 112, 107805, 2021.
- Hashemi, H. and Shaharoun, A.M., *A case-based reasoning approach for design of machining fixture*, International Journal of Advanced Manufacturing Technology, 74, 113–124, 2014.
- Hodge, V. and Austin, J., *A survey of outlier detection methodologies*, Artificial Intelligence Review, 22, 85–126, 2004.
- Hu, J., Qi, J., and Peng, Y., *New case-based reasoning adaptation method combining with problem-solution relational analysis for mechanical design*, Computers in Industry, 66, 41–51, 2015.
- Hung, Y.H., *A neural network classifier with rough set-based feature selection to classify multiclass IC package products*, Advanced Engineering Informatics, 23(3), 348–357, 2009.
- Jung, S., Lim, T., and Kim, D., *Integrating radial basis function networks with case-based reasoning for product design*, Expert Systems With Applications, 36, 5695–5701, 2009.
- Kumru, M. and Kumru, P., *Using artificial neural networks to forecast operation times in metal industry*, International Journal of Computer Integrated Manufacturing, 27, 48–59, 2014.
- Liu, C., Chang, P., and Kao, I., *Cluster-based evolving FCBR for flow time prediction in semiconductor manufacturing factory*, Proceedings of the 8th WSEAS International Conference on Applied Computer Science (ACS'08), 21–23, Venice, Italy, 2008.
- Mourtzis, D., Doukas, M., Fragou, K., Efthymiou, K., and Matzorou, V., *Knowledge-based estimation of manufacturing lead time for complex engineered-to-order products*, Procedia CIRP, 17, 499–504, 2014.
- Omuya, E.O., Okeyo, G.O., and Kimwele, M.W., *Feature selection for classification using principal component analysis and information gain*, Expert Systems with Applications, 174, 114765, 2021.
- Sample, N., Haines, M., Arnold, M., and Purcell, T., *Optimizing search strategies in kd trees*, Fifth WSES/IEEE World Multiconference on Circuits, Systems, Communications & Computers (CSCC 2001), 2001.

- Sigweni, B. and Shepperd, M., *Feature weighting techniques for CBR in software effort estimation studies : A review and empirical evaluation*, Proceedings of the 10th International Conference on Predictive Models in Software Engineering, 32–41, Turin, Italy, September 2014.
- Singh, K. and Upadhyaya, S., *Outlier detection : applications and techniques*, International Journal of Computer Science Issues (IJCSI), 9(1), 307, 2012.
- Sylla, A., Coudert, T., and Geneste, L., *A case-based reasoning approach for engineer-to-order systems performance evaluation*, IFAC-PapersOnLine, 54(1), 717–722, 2021.
- Sylla, A., Guillon, D., Vareilles, E., Aldanondo, M., Coudert, T., and Geneste, L., *Configuration knowledge modeling : How to extend configuration from assemble/make to order towards engineer to order for the bidding process*.Computers in Industry, 99, 29–41, 2018.
- Szczepaniak, P.S. and Duraj, A., *Case-based reasoning : The search for similar solutions and identification of outliers*, Complexity, 1–12, 2018.
- Shah, K., Kopru, S., Ruvini, J. D., *Neural network based extreme classification and similarity models for product matching*, Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics : Human Language Technologies, 8–15, 2018.
- Zhu, B. and Jiang, P., *An approach to configuring product family using rough set theory*, International Journal of Product Development, 2(1-2), 155-169, 2005.
- He, C., Li, Z.K., Wang, S., and Liu, D.Z., *A systematic data-mining-based methodology for product family design and product configuration*, Advanced Engineering Informatics, 48, 101302, 2021.
- Mansoul, A. and Atmani, B., *Clustering to enhance case-based reasoning*, In Modelling and Implementation of Complex Systems : Proceedings of the 4th International Symposium, MISC 2016, Constantine, Algeria, May 7-8, 2016, Constantine, Algeria (pp. 137-151), Springer International Publishing, 2016.
- Zhang, Q., Shi, C., Niu, Z., and Cao, L., *HCBC : A Hierarchical Case-Based Classifier Integrated with Conceptual Clustering*, IEEE Transactions on Knowledge and Data Engineering, 31(1), 152-165, 2019.
- Müller, G. and Bergmann, R., *A cluster-based approach to improve similarity-based retrieval for process-oriented case-based reasoning*, In ECAI 2014 (pp. 639-644), IOS Press, 2014.
- Kim, K. S. and Han, I., *The cluster-indexing method for case-based reasoning using self-organizing maps and learning vector quantization for bond rating cases*, Expert systems with applications, 21(3), 147-156, 2001.
- Yang, Q. and Wu, J., *Enhancing the effectiveness of interactive case-based reasoning with clustering and decision forests*, Applied Intelligence, 14, 49-64, 2001.
- Perez, B., Lang, C., Henriet, J., Philippe, L., and Auber, F., *Risk prediction in surgery using case-based reasoning and agent-based modelization*, Computers in Biology and Medicine, 2021, 128, 104040, 2021.
- Li, S. M., and Lee, C. K. M., *Transfer-Learning-Based Opinion Mining for New-Product Portfolio Configuration over the Case-Based Reasoning Cycle*, Applied Sciences, 2022, 12(23), 12477, 2022.
- Ma'ruf, A., Thoriq, D.A., & Buwana, K.S., *An Early Machining Time Estimation for Make-to-Order Manufacturing Using Machine Learning Approach*, Procedia CIRP, 130, 106–111, 2024.
- Yamashiro, H., & Nonaka, H., *Estimation of processing time using machine learning and real factory data for optimization of parallel machine scheduling problem*, Operations Research Perspectives, 8, 100196, 2021.
- Cotacallapa, H., Saboya, N., Rodrigues, P.C., Salas, R., & López-Gonzales, J.L., *A flat-hierarchical approach based on machine learning model for e-commerce product classification*, IEEE Access, 2024.
- Roberson, A., *Applying machine learning for automatic product categorization*, Journal of Official Statistics, 37(2), 395–410, 2021.